

Ensembles

Combine models together

bagging

boosting

What can be an ensemble?

- Different models

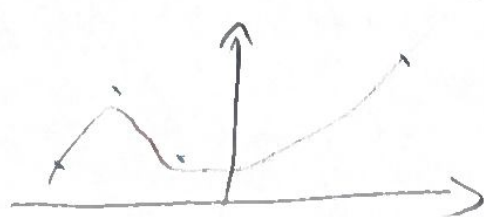
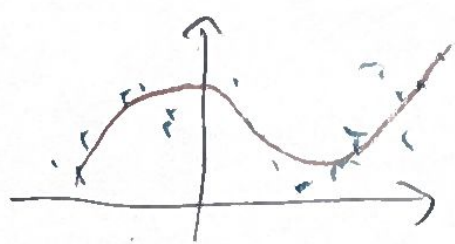
- Different choice of hyperparam.

- Different training data

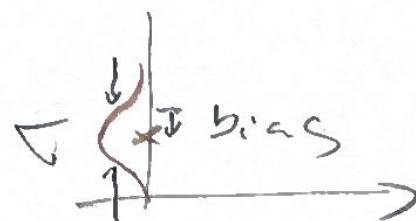
- Training data with different weight

Bagging: train using "independent" random subsets of data

Boosting: sequential training of classifiers with focus on misclassified data

Bias and variance of ensemble

Ensemble



12

Independent training data, (m)

$$h = \frac{1}{m} \sum_{i=1}^m h_i$$

- Average unchanged (bias)

$$E[h(x)] = \frac{1}{m} \sum_i E[h_i(x)]$$

- Variance reduced

$$\begin{aligned} \text{Var}[h(x)] &= \text{Var}\left[\frac{1}{m} \sum_i h_i(x)\right] = \\ &= \frac{1}{m^2} \sum_i \text{Var}[h_i(x)] = \frac{1}{m} \text{Var}[h_i(x)] \end{aligned}$$

Independent dataset, is expensive/difficult

Bootstrap aggregation or bagging

- Dataset D with n examples

- m new datasets D_i by replacement sampling of D

- Average model results

Problem: data is not independent

ρ : average correlation between D_i

$$\text{Var}\left[\frac{1}{m} \sum_i h_i(x)\right] = \frac{1}{m} (1-\rho) \sigma^2 + \rho \sigma^2$$

Random forest: bagged decision trees

- bagged data
- at each branch only a random subset of features

Parameter importance

- At each D_i there are out of bag data points
- Measure error of these
- Shuffle one feature for these and measure error again
- Larger increase in error means more important feature (param)

Random forest is the best black box method for up to a few dozen params.

4

Boosting: Weighted training set

$$\sum_i w^{(i)} = 1 \quad w^{(i)} > 0$$

$$Err^{(w)} = \sum_i w^{(i)} I[h(x^{(i)}) \neq y^{(i)}]$$

Weak learner: $Err = \frac{1}{2} - \gamma$ $\gamma < 1$

e.g. low depth Decision tree
(can even be 1 branch)

Adaboost

1. At each iteration (f) reweight training data by assigning larger weight to misclassified data
2. Train classifier again
3. Add classifier to ensemble with appropriate weight
4. Go to 1.

For

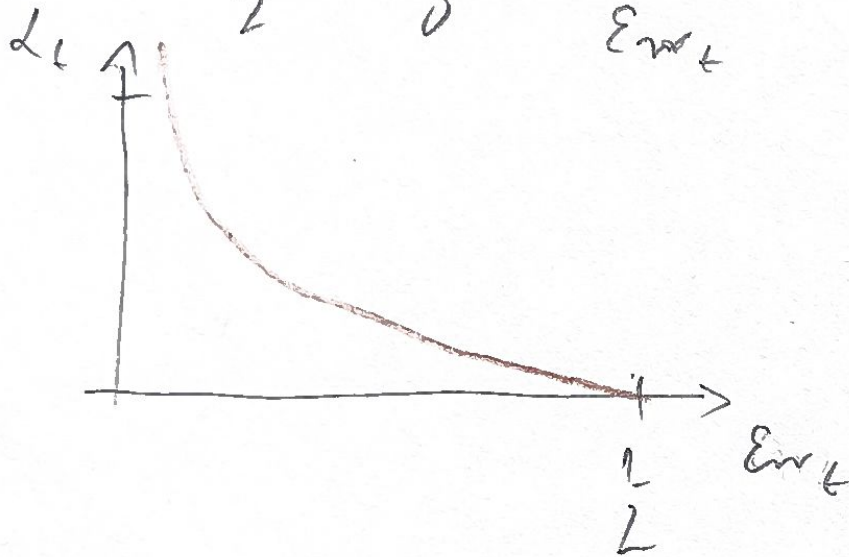
Ada boost

For simplicity: $y^{(i)} = \pm 1$

Classifier coefficient: α_t

- high for good performing ones

$$\alpha_t = \frac{1}{2} \log \frac{1 - \text{Err}_t}{\text{Err}_t}$$

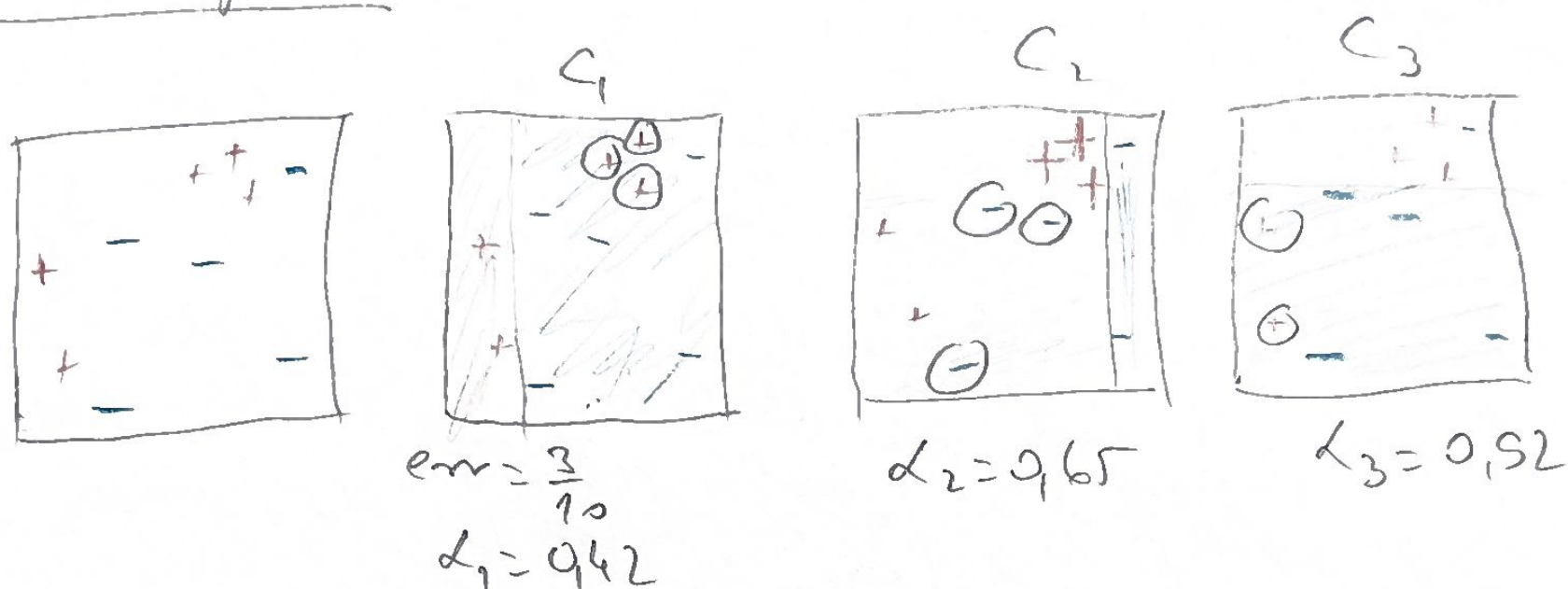


$$w_{t+1}^{(i)} = w_t^{(i)} \cdot \exp(-\alpha_t \cdot y^{(i)} h_t(x^{(i)}))$$

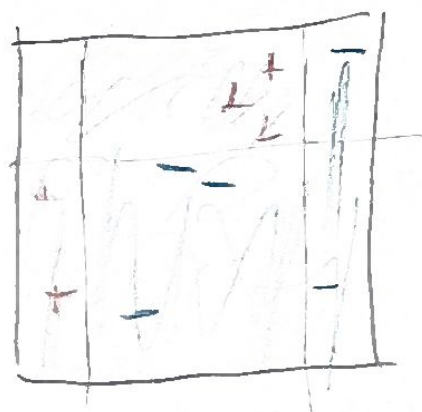
Classification: $\rightarrow$ $\sum \text{sgn } h = \text{sgn } y \exp(-\dots)$
 $\rightarrow$ $\text{is } \text{sgn } h \neq \text{sgn } y \exp(+\dots)$

Example

6



$$C = d_1 C_1 + d_2 C_2 + d_3 C_3$$



Summary

Bagging:

- Reduces variance
- Bias not changed
- Robust to outliers
- Reduces overfitting
- Parallel

Boosting

- Reduces bias
- Increase variance
- May overfit

- sequential